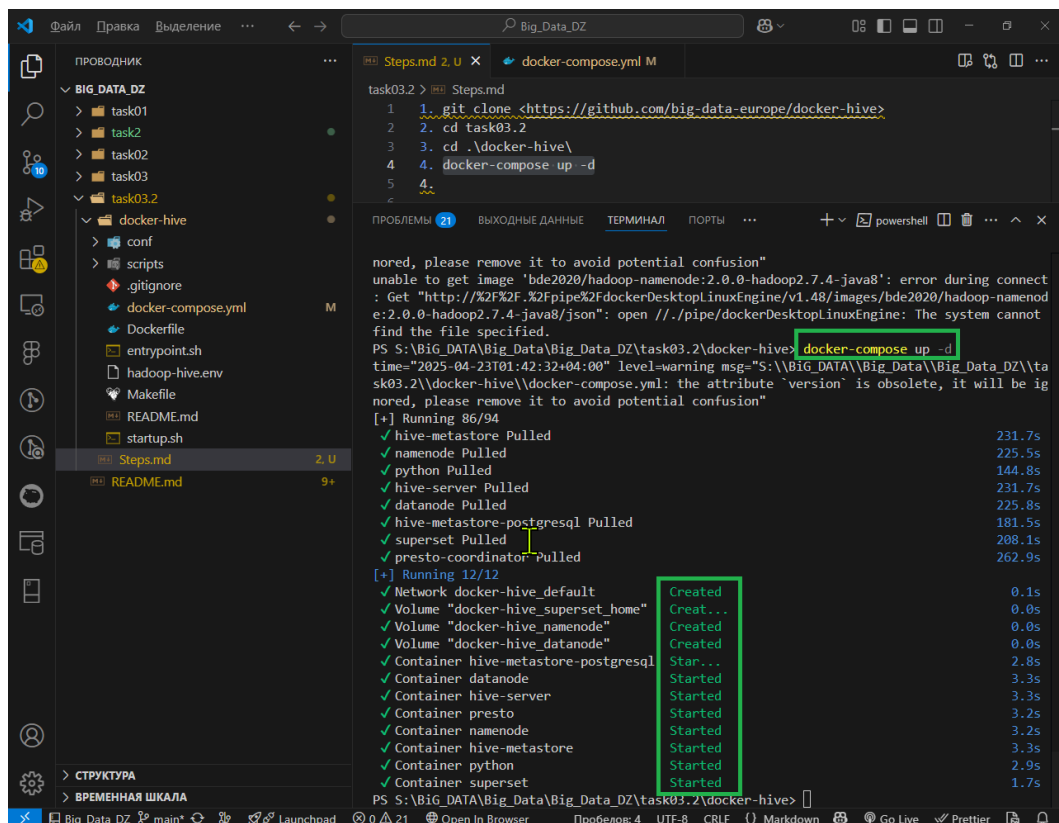
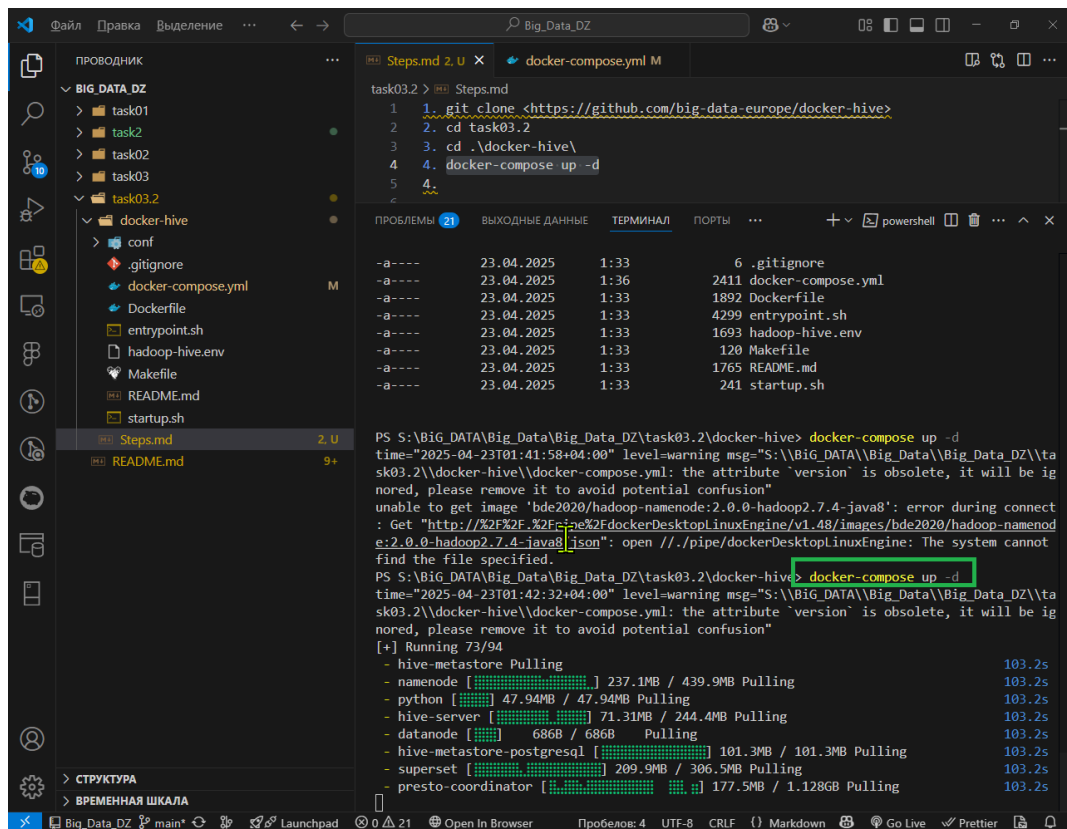
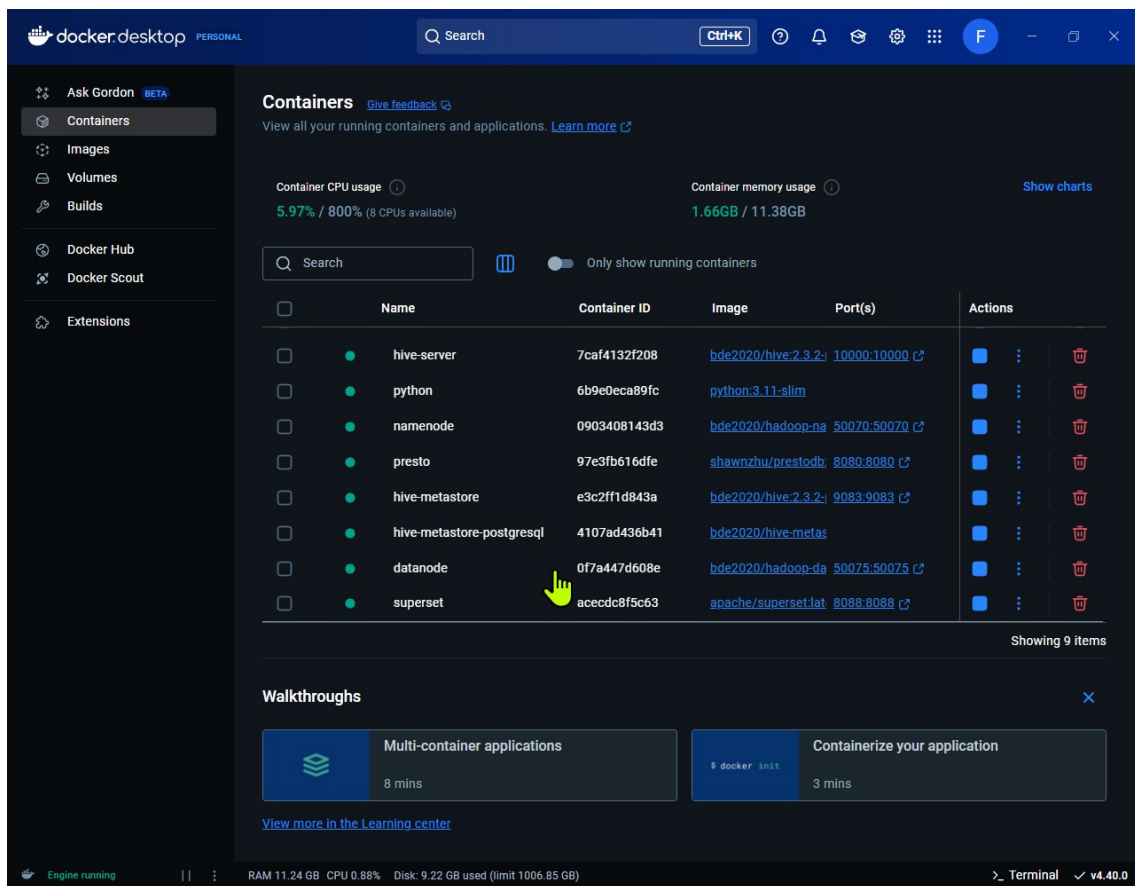
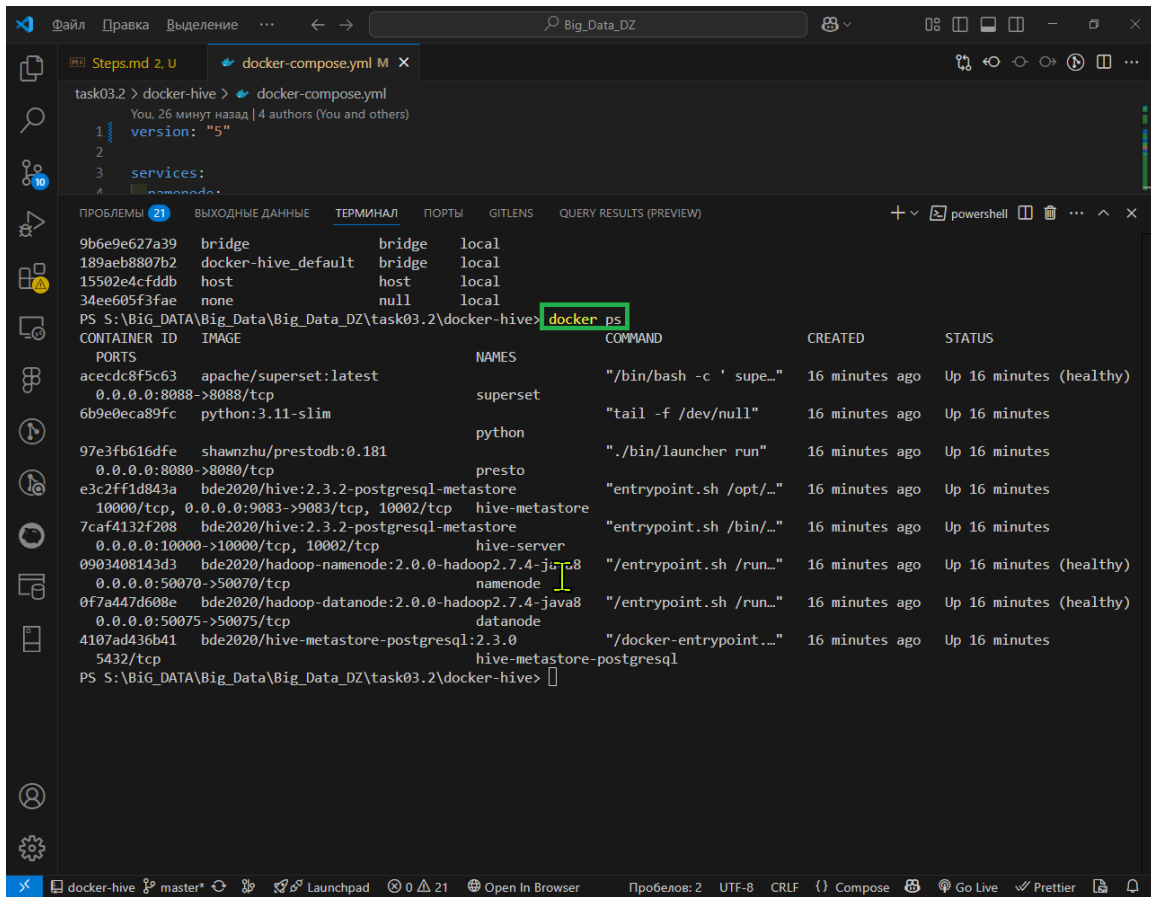
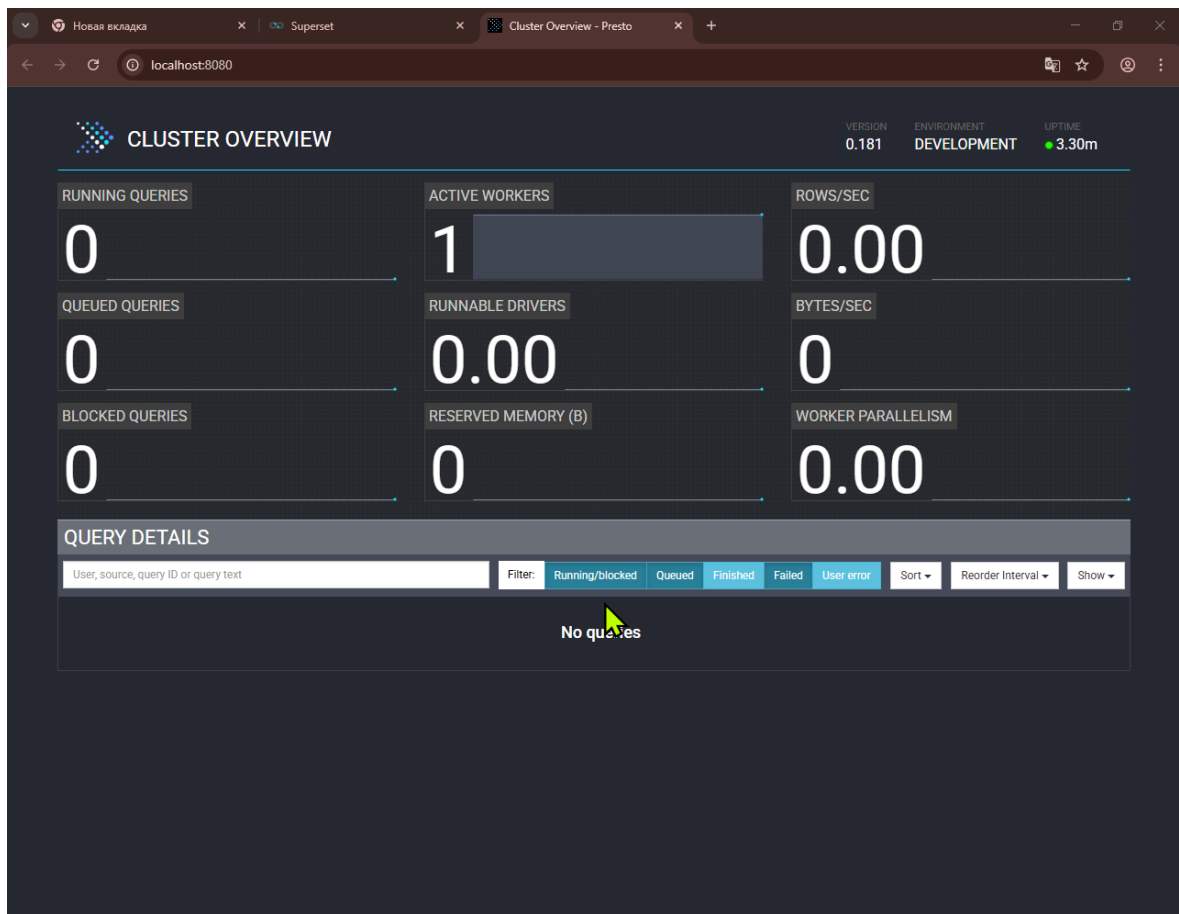








Superset Settings • Login

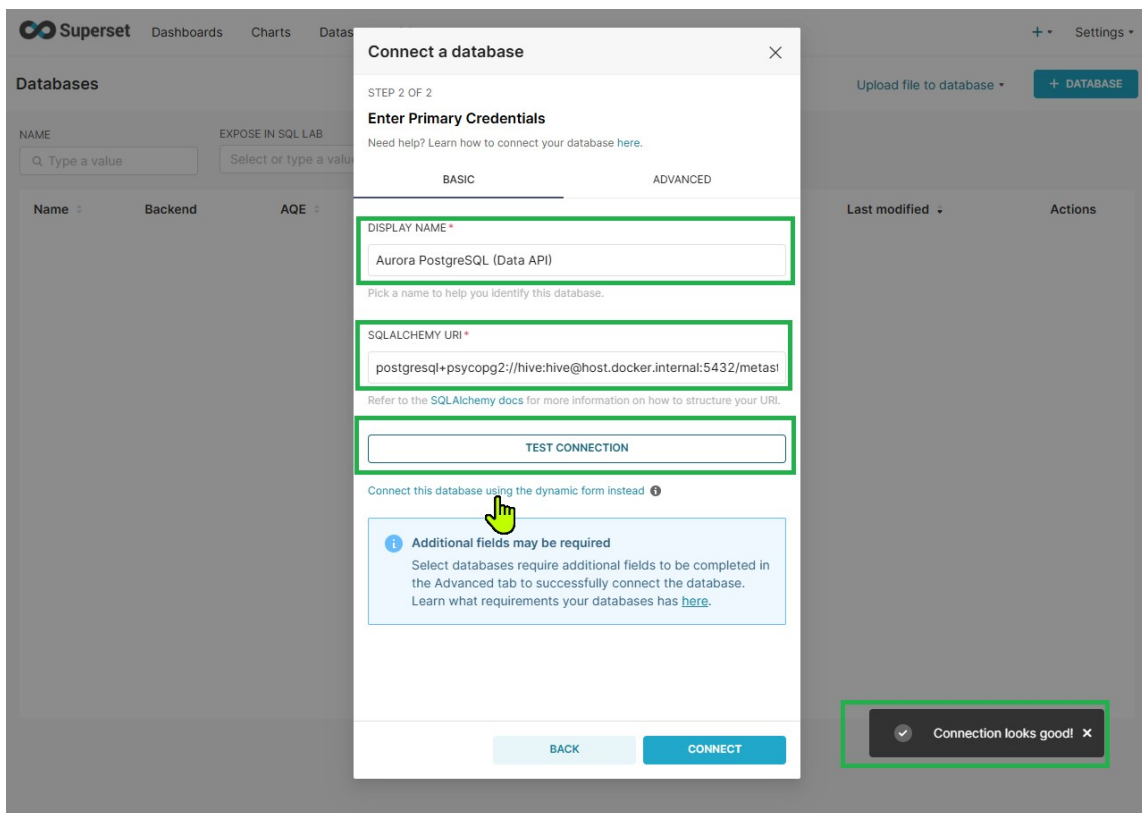
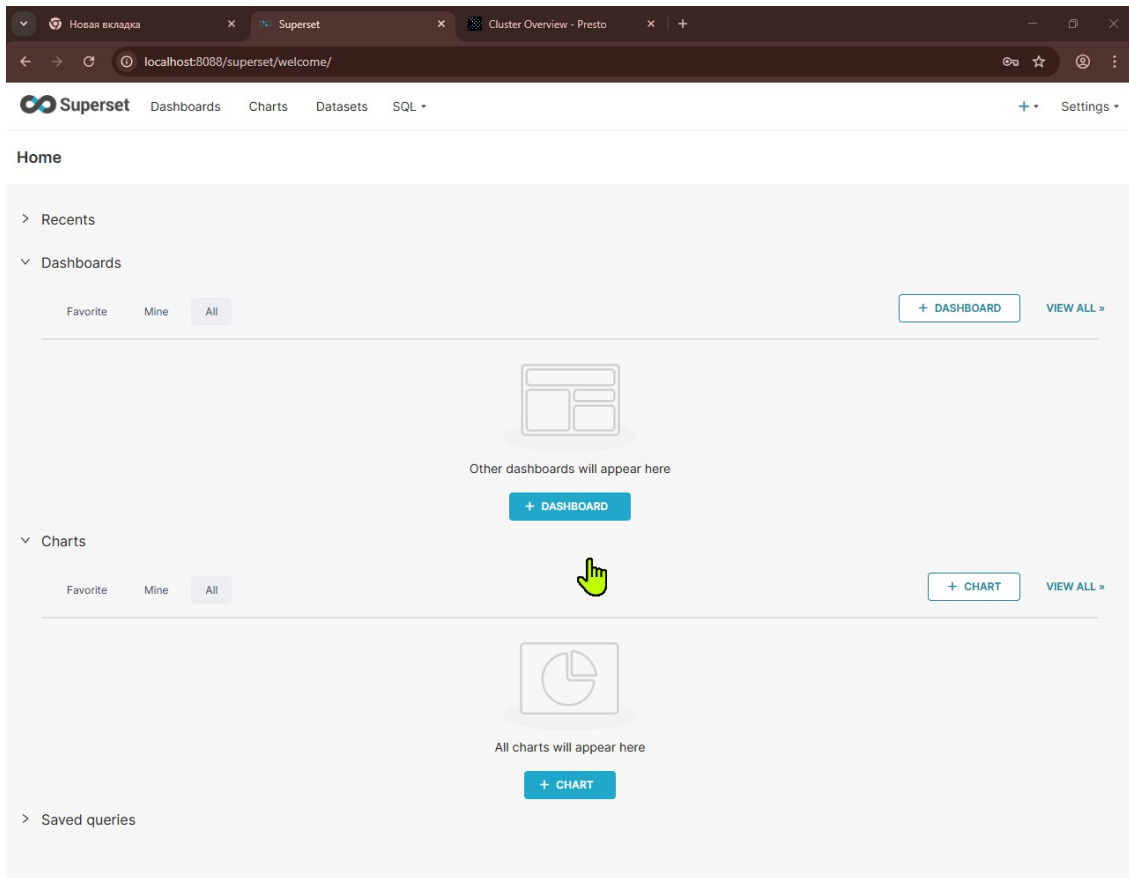
Sign In

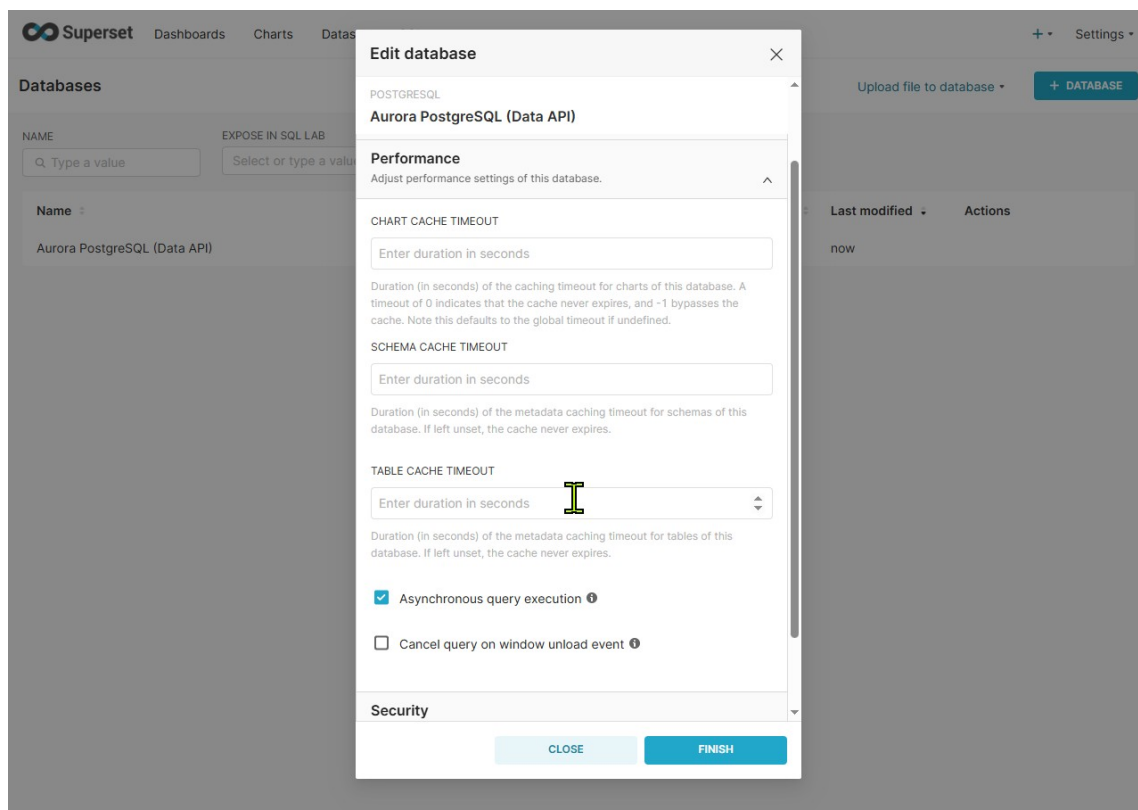
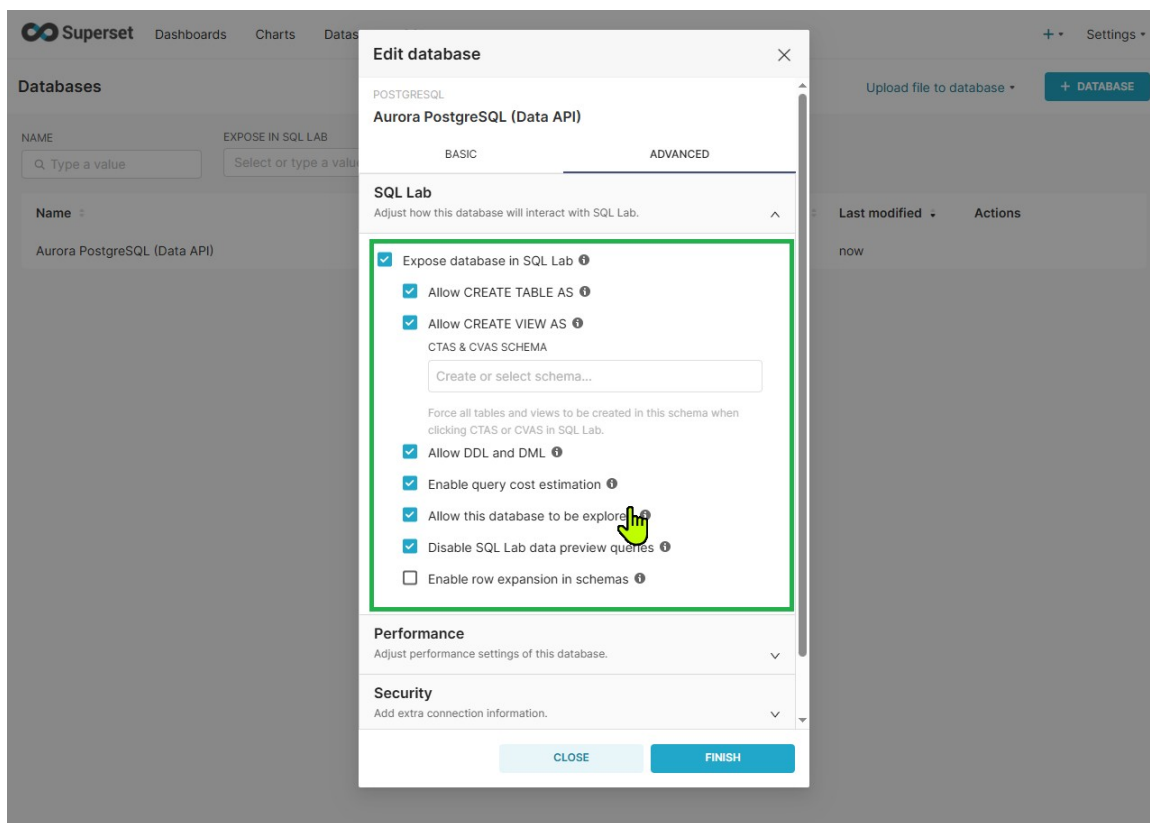
Enter your login and password below:

USERNAME: admin

PASSWORD:

SIGN IN





Databases

Upload file to database

+ DATABASE

NAME EXPOSE IN SQL LAB AQE MODIFIED BY

Name	Backend	AQE	DML	File upload	Expose in SQL Lab	Last modified	Actions
Aurora PostgreSQL (Data API)	postgresql	✓	✓	✗	✓	now	

« 1 »
1-1 of 1



Databases

Upload file to database

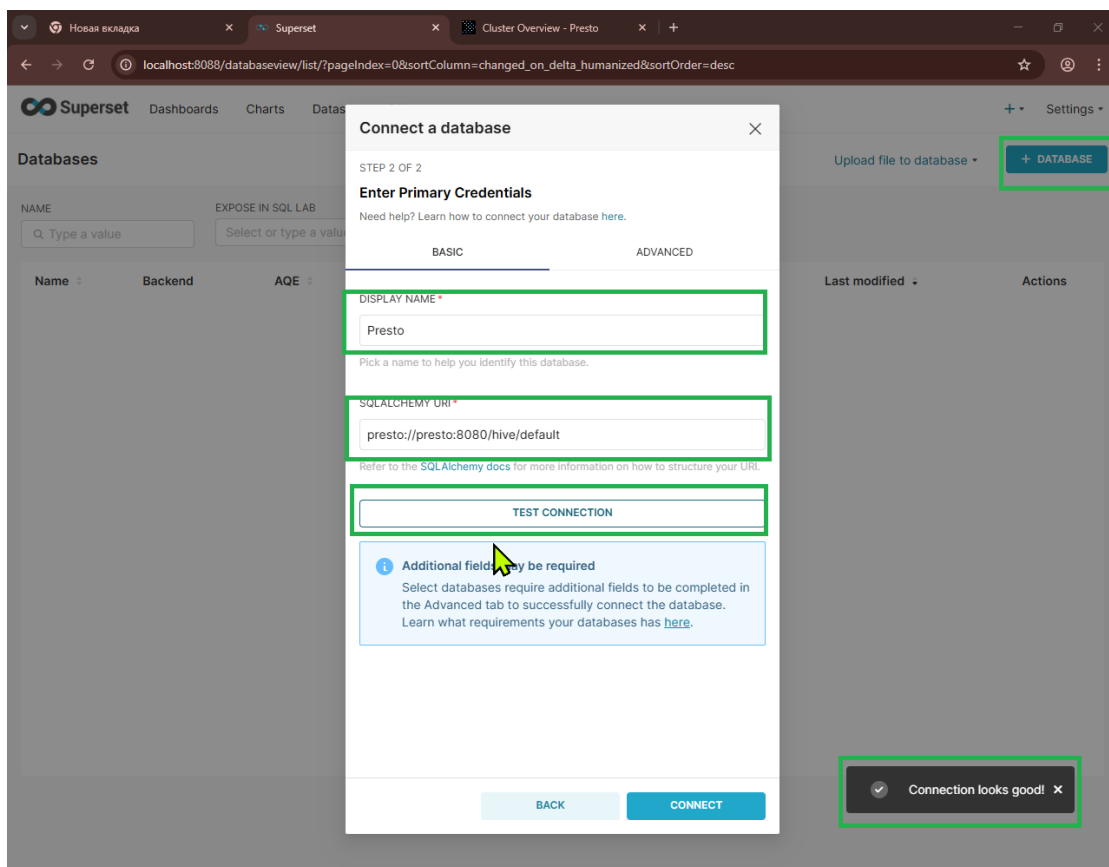
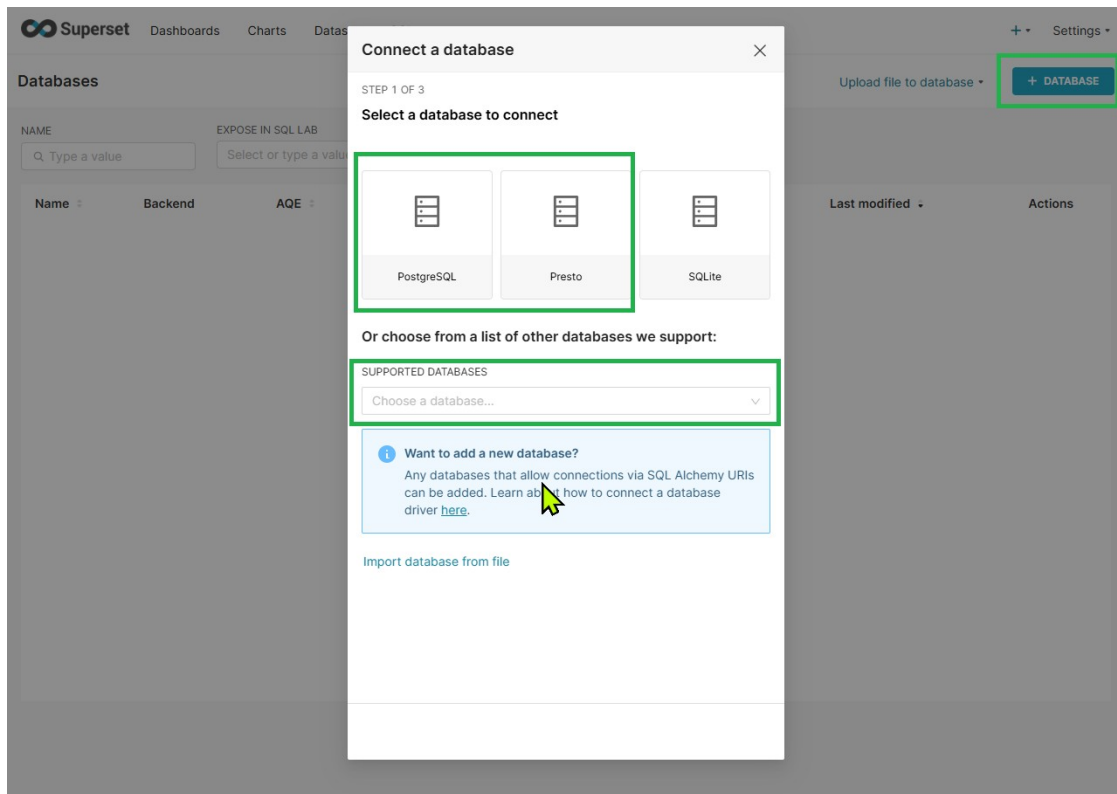
+ DATABASE

NAME EXPOSE IN SQL LAB AQE MODIFIED BY

Name	Backend	AQE	DML	File upload	Expose in SQL Lab	Last modified	Actions
Aurora PostgreSQL (Data API)	postgresql	✗	✓	✓	✓	now	

« 1 »
1-1 of 1





Superset Dashboards Charts Datasets SQL + Settings +

Databases

Upload file to database + DATABASE

Name	Expose in SQL Lab	AQE	DML	File upload	Expose in SQL Lab	Last modified	Actions
Aurora PostgreSQL (Data API)	postgresql	✓	✓	✓	✓	now	
Presto	presto	✓	✓	✓	✓	a minute ago	

« 1 »
1-2 of 2

Файл Правка Выделение ... Big_Data_DZ

task03.2 > Steps.md

```
17 14. SQLAlchemy URI: presto://presto:8080/hive/default
18 15. SQLAlchemy URI: hive://hive-server:10000/default
19 16. ./scripts/weather_fetcher.py
20 17. docker exec -it python python weather_fetcher.py
21 18. docker exec -it python bash
22 19. apt update && apt install -y python3-pip
23 20. pip install requests pandas
24 21.
...
```

ПРОБЛЕМЫ 14 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS ... + docker

```
import requests
ModuleNotFoundError: No module named 'requests'
root@52284c6f2a4:/# pip install requests pandas
Collecting requests
  Downloading requests-2.32.3-py3-none-any.whl.metadata (4.6 kB)
Collecting pandas
  Downloading pandas-2.2.3-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (89 kB)
  89.9/89.9 kB 821.7 kB/s eta 0:00:00
Collecting charset-normalizer<4,>=2 (from requests)
  Downloading charset_normalizer-3.4.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.m
  etadata (35 kB)
Collecting idna<4,>=2.5 (from requests)
  Downloading idna-3.10-py3-none-any.whl.metadata (10 kB)
Collecting urllib3<3,>=1.21.1 (from requests)
  Downloading urllib3-2.4.0-py3-none-any.whl.metadata (6.5 kB)
Collecting certifi>=2017.4.17 (from requests)
  Downloading certifi-2025.1.31-py3-none-any.whl.metadata (2.5 kB)
Collecting numpy>=1.23.2 (from pandas)
  Downloading numpy-2.2.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (62 k
  B)
  62.0/62.0 kB 2.4 MB/s eta 0:00:00
Collecting python-dateutil>=2.8.2 (from pandas)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting pytz>=2020.1 (from pandas)
  Downloading pytz-2025.2-py2.py3-none-any.whl.metadata (22 kB)
Collecting tzdata>=2022.7 (from pandas)
  Downloading tzdata-2025.2-py2.py3-none-any.whl.metadata (1.4 kB)
Collecting six>=1.5 (from python-dateutil>=2.8.2->pandas)
```


The screenshot shows the VS Code editor with a file explorer on the left and a terminal on the right. The file explorer shows a project named 'BIG_DATA_DZ' with a subdirectory 'task03.2' containing a 'docker-hive' directory. The terminal shows the execution of a script 'Steps.md' which runs 'docker exec -it python python weather_fetcher.py'. The output shows the installation of various Python packages using pip, including certifi, charset-normalizer, idna, numpy, pandas, python-dateutil, pytz, requests, six, tzdata, and urllib3. A warning message is displayed: 'WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv'. The terminal also shows the execution of 'python /scripts/weather_fetcher.py' and the saving of data to 'weather_data.csv'.

```
task03.2 > Steps.md
17 14. SQLAlchemy URI: presto://presto:8080/hive/default
18 15. SQLAlchemy URI: hive://hive-server:10000/default
19 16. /scripts/weather_fetcher.py
20 17. docker exec -it python python weather_fetcher.py
21 18. docker exec -it python bash
22 19. apt update && apt install -y python3-pip
23 20. pip install requests pandas
24 21.

PROБЛЕМЫ 14 Выходные данные ТЕРМИНАЛ ПОРТЫ GITLENS ...
Downloading certifi-2025.1.31-py3-none-any.whl (166 kB)
13.1/13.1 MB 8.1 MB/s eta 0:00:00
Downloading charset-normalizer-3.4.1-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (143 kB)
166.4/166.4 kB 4.3 MB/s eta 0:00:00
Downloading idna-3.10-py3-none-any.whl (70 kB)
143.9/143.9 kB 8.2 MB/s eta 0:00:00
Downloading numpy-2.2.5-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (16.4 MB)
70.4/70.4 kB 6.7 MB/s eta 0:00:00
Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl (229 kB)
16.4/16.4 MB 8.6 MB/s eta 0:00:00
Downloading pytz-2025.2-py3-none-any.whl (509 kB)
229.9/229.9 kB 8.5 MB/s eta 0:00:00
Downloading tzdata-2025.2-py2.py3-none-any.whl (347 kB)
509.2/509.2 kB 10.9 MB/s eta 0:00:00
Downloading urllib3-2.4.0-py3-none-any.whl (128 kB)
347.8/347.8 kB 9.4 MB/s eta 0:00:00
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
128.7/128.7 kB 5.7 MB/s eta 0:00:00
Installing collected packages: pytz, urllib3, tzdata, six, numpy, idna, charset-normalizer, certifi, requests, python-dateutil, pandas
Successfully installed certifi-2025.1.31 charset-normalizer-3.4.1 idna-3.10 numpy-2.2.5 pandas-2.2.3 python-dateutil-2.9.0.post0 pytz-2025.2 requests-2.32.3 six-1.17.0 tzdata-2025.2 urllib3-2.4.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
root@52284c6ef2a4:/# python /scripts/weather_fetcher.py
Данные сохранены в weather_data.csv
root@52284c6ef2a4:/#
```

The screenshot shows the VS Code editor with a file explorer on the left and a terminal on the right. The file explorer shows the 'task03.2' directory with a 'docker-hive' subdirectory. The terminal shows the execution of 'python /scripts/weather_fetcher.py' which outputs a JSON object representing weather data for London. The output is:

```
{
  "longitude": lon,
  "start_date": start_date,
  "end_date": end_date,
  "daily": "temperature_2m_max,temperature_2m_min",
  "timezone": "UTC"
}
```

. The terminal also shows the execution of 'ls' and 'ls -la' commands, and the resulting output of the 'ls' command:

```
total 8
drwxrwxrwx 1 root root 512 Apr 22 23:42 .
drwxr-xr-x 1 root root 4096 Apr 23 00:04 ..
-rwxrwxrwx 1 root root 1047 Apr 22 23:47 weather_fetcher.py
```

. The terminal also shows the execution of 'cd ..' and 'ls' commands, and the resulting output of the 'ls' command:

```
bin dev home lib64 mnt proc run scripts sys usr
boot etc lib media opt root sbin srv tmp var
```

. The terminal also shows the execution of 'cd /scripts/' and 'ls' commands, and the resulting output of the 'ls' command:

```
weather_fetcher.py
```

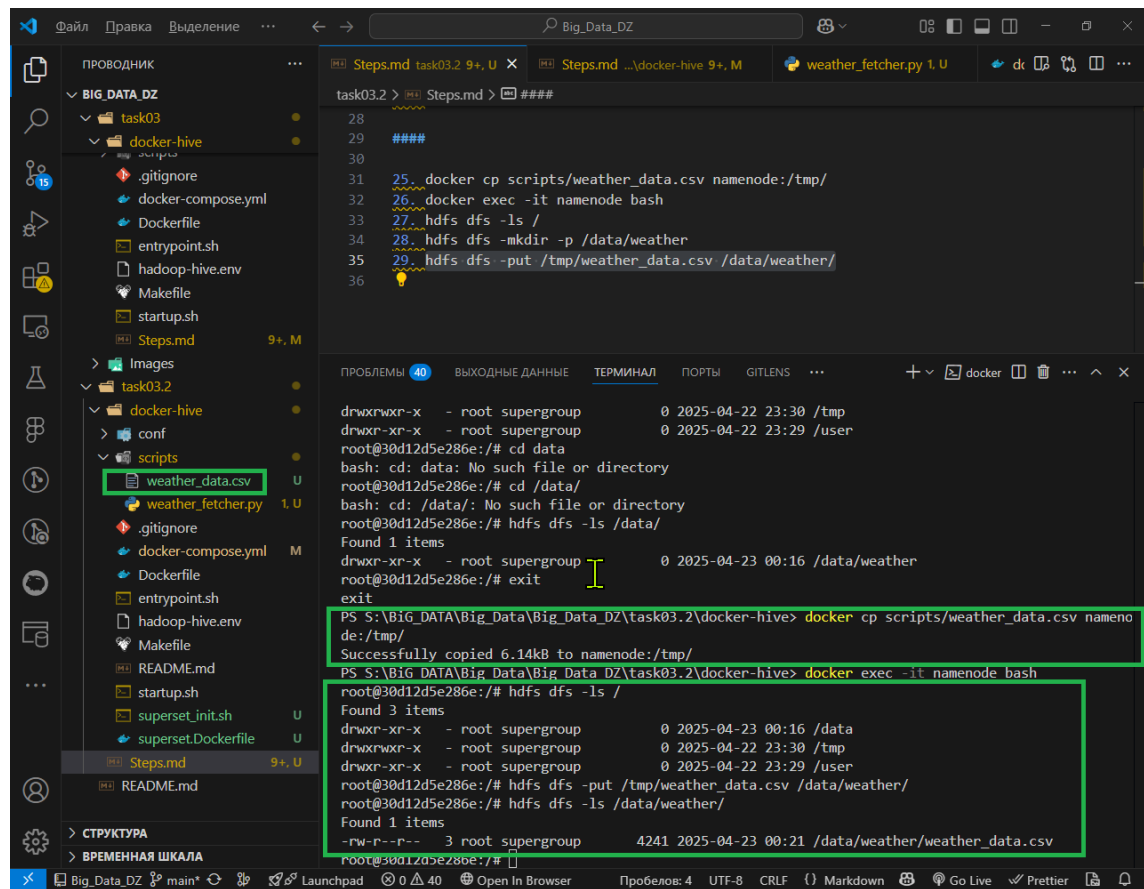
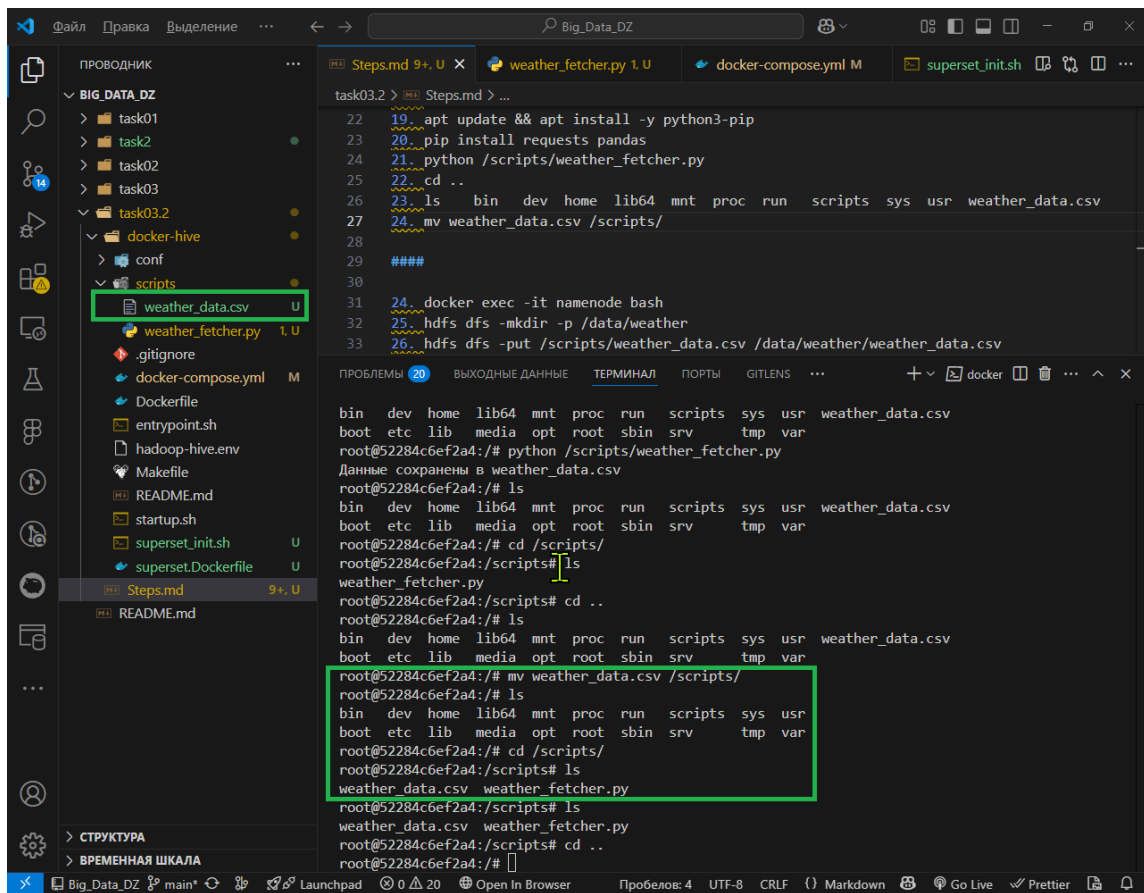
. The terminal also shows the execution of 'cd ..' and 'ls' commands, and the resulting output of the 'ls' command:

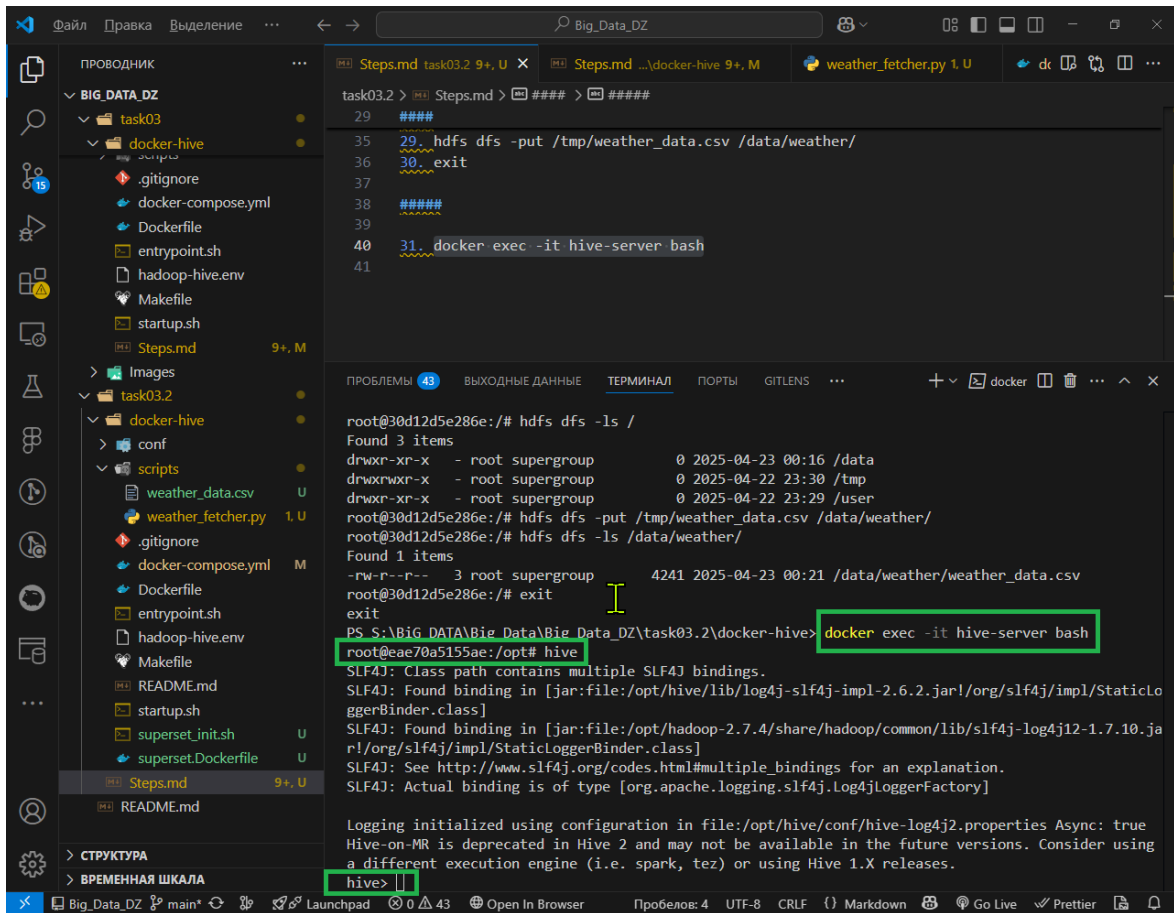
```
weather_data.csv
```

.

```
task03.2 > docker-hive > scripts > weather_fetcher.py ...
23 "longitude": lon,
24 "start_date": start_date,
25 "end_date": end_date,
26 "daily": "temperature_2m_max,temperature_2m_min",
27 "timezone": "UTC"
28 }
29 r = requests.get(url, params=params)
30 json_data = r.json()
31 df = pd.DataFrame(json_data["daily"])
32 df["city"] = city
33 all_data.append(df)
34

PROБЛЕМЫ 14 Выходные данные ТЕРМИНАЛ ПОРТЫ GITLENS ...
Downloading urllib3-2.4.0-py3-none-any.whl (128 kB)
128.7/128.7 kB 5.7 MB/s eta 0:00:00
Downloading six-1.17.0-py2.py3-none-any.whl (11 kB)
Installing collected packages: pytz, urllib3, tzdata, six, numpy, idna, charset-normalizer, certifi, requests, python-dateutil, pandas
Successfully installed certifi-2025.1.31 charset-normalizer-3.4.1 idna-3.10 numpy-2.2.5 pandas-2.2.3 python-dateutil-2.9.0.post0 pytz-2025.2 requests-2.32.3 six-1.17.0 tzdata-2025.2 urllib3-2.4.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
root@52284c6ef2a4:/# python /scripts/weather_fetcher.py
Данные сохранены в weather_data.csv
root@52284c6ef2a4:/# cd /scripts/
root@52284c6ef2a4:/scripts# ls
weather_fetcher.py
root@52284c6ef2a4:/scripts# ls -la
total 8
drwxrwxrwx 1 root root 512 Apr 22 23:42 .
drwxr-xr-x 1 root root 4096 Apr 23 00:04 ..
-rwxrwxrwx 1 root root 1047 Apr 22 23:47 weather_fetcher.py
root@52284c6ef2a4:/scripts# cd ..
root@52284c6ef2a4:/# ls
bin dev home lib64 mnt proc run scripts sys usr
boot etc lib media opt root sbin srv tmp var
root@52284c6ef2a4:/#
root@52284c6ef2a4:/# cd /scripts/
root@52284c6ef2a4:/scripts# ls
weather_fetcher.py
root@52284c6ef2a4:/scripts# cd ..
root@52284c6ef2a4:/# ls
weather_data.csv
root@52284c6ef2a4:/#
```





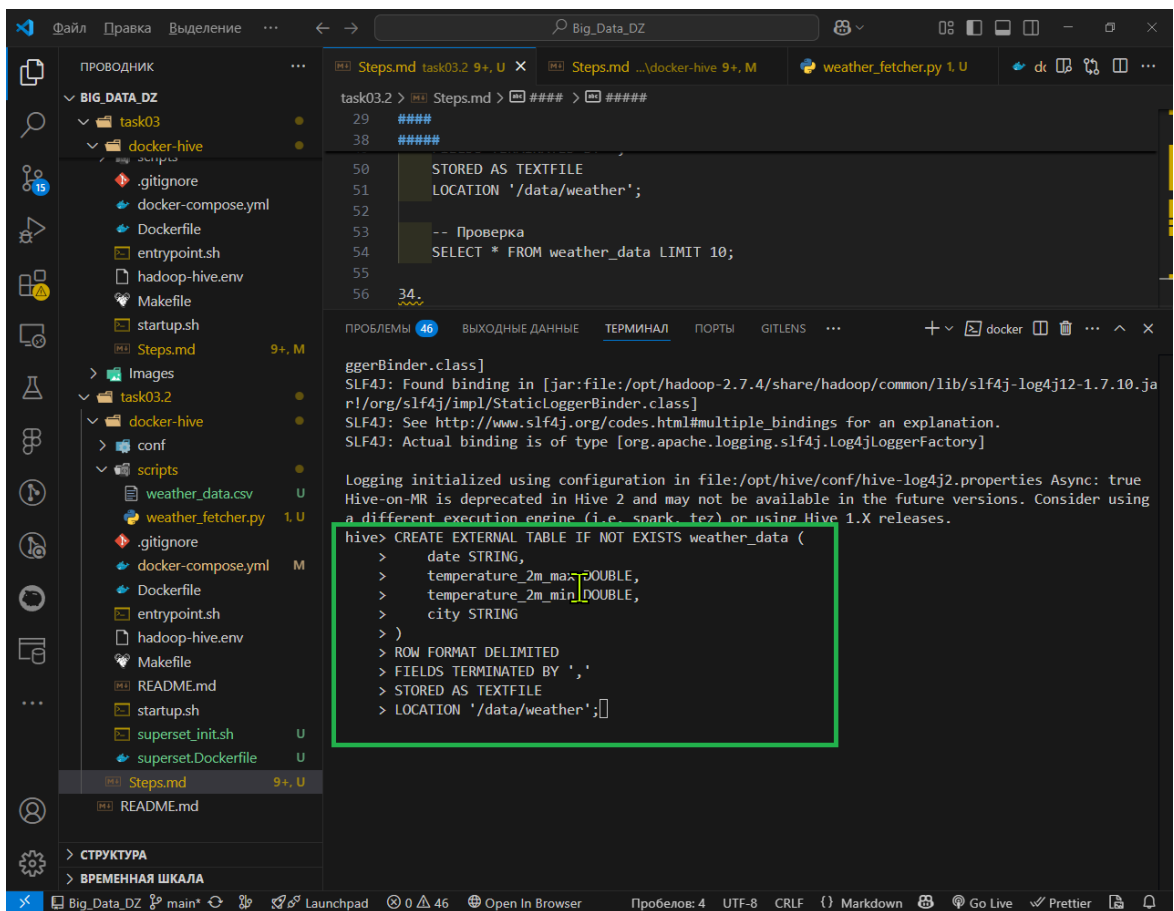
```
task03.2 > Steps.md > ##### > #####
29 #####
35 29. hdfs dfs -put /tmp/weather_data.csv /data/weather/
36 30. exit
37
38 #####
39
40 31. docker exec -it hive-server bash
41
```

PROБЛЕМЫ 43 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS ...

```
root@30d12d5e286e:/# hdfs dfs -ls /
Found 3 items
drwxr-xr-x - root supergroup 0 2025-04-23 00:16 /data
drwxrwxr-x - root supergroup 0 2025-04-22 23:30 /tmp
drwxr-xr-x - root supergroup 0 2025-04-22 23:29 /user
root@30d12d5e286e:/# hdfs dfs -put /tmp/weather_data.csv /data/weather/
root@30d12d5e286e:/# hdfs dfs -ls /data/weather/
Found 1 items
-rw-r--r-- 3 root supergroup 4241 2025-04-23 00:21 /data/weather/weather_data.csv
root@30d12d5e286e:/# exit
exit
PS S:\Big Data\Big Data DZ\task03.2\docker-hive> docker exec -it hive-server bash
root@eae70a5155ae:/opt# hive
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/hive/lib/log4j-slf4j-impl-2.6.2.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop-2.7.4/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]

Logging initialized using configuration in file:/opt/hive/conf/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.

hive>
```



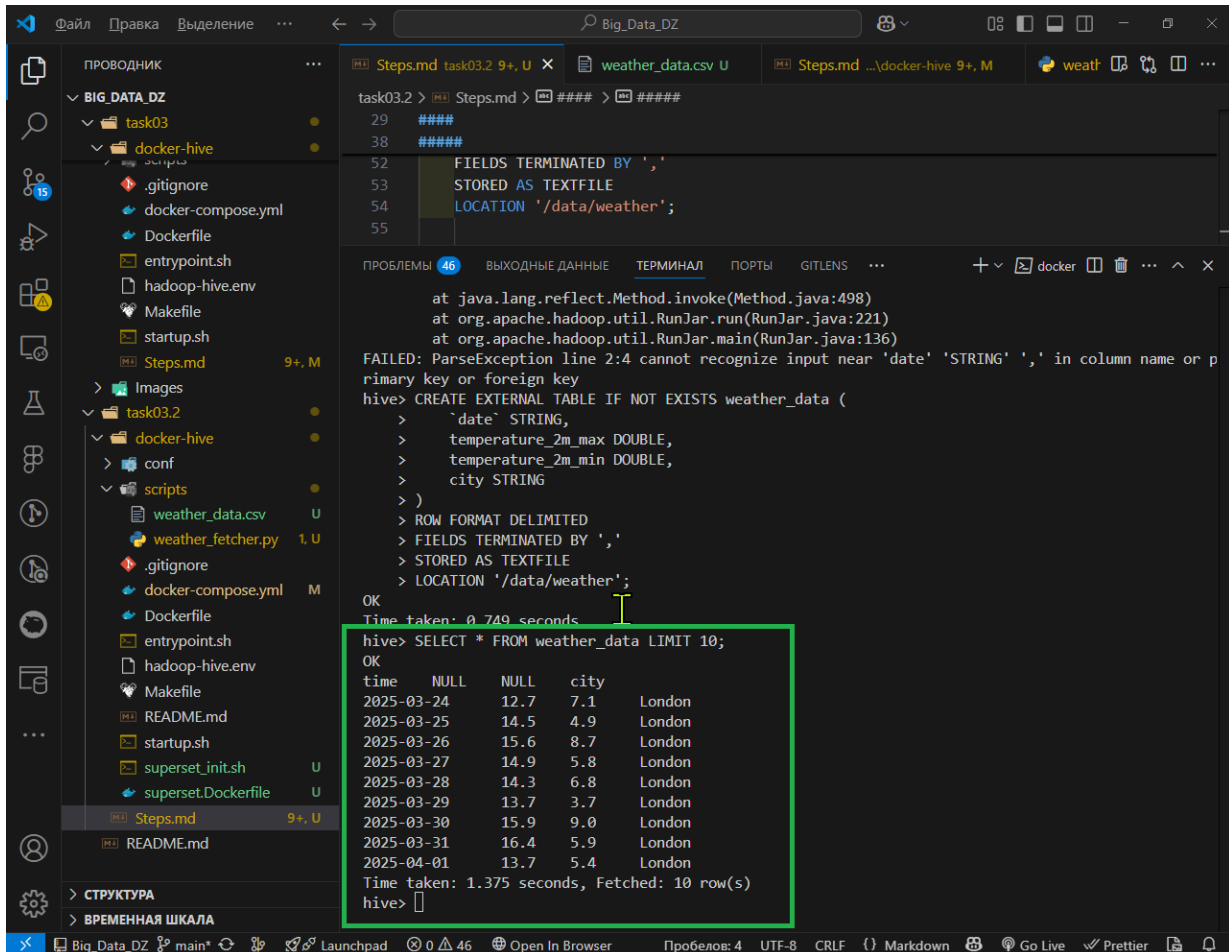
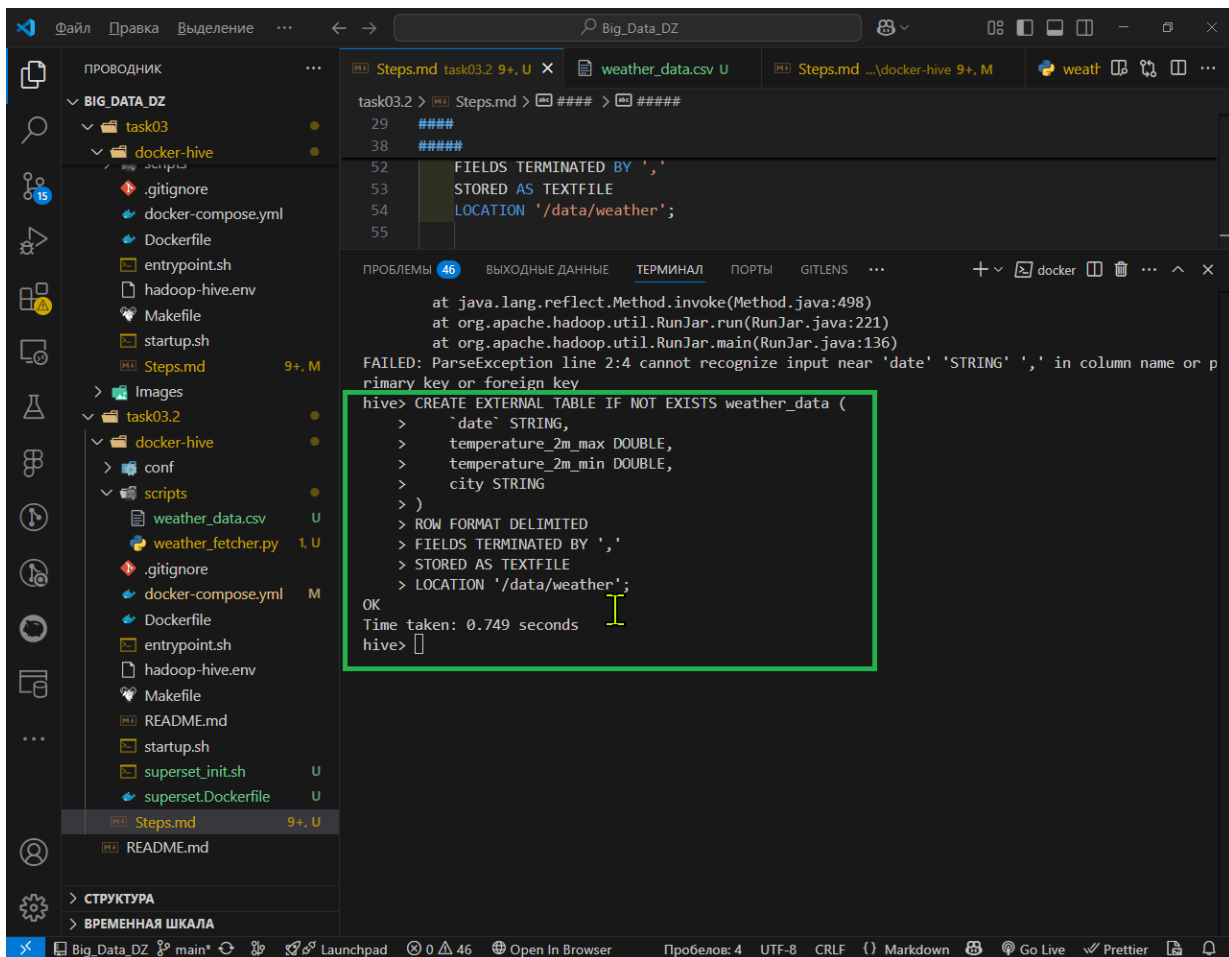
```
task03.2 > Steps.md > ##### > #####
29 #####
38 #####
50 STORED AS TEXTFILE
51 LOCATION '/data/weather';
52
53 -- Проверка
54 SELECT * FROM weather_data LIMIT 10;
55
56 34.
```

PROБЛЕМЫ 46 ВЫХОДНЫЕ ДАННЫЕ ТЕРМИНАЛ ПОРТЫ GITLENS ...

```
ggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop-2.7.4/share/hadoop/common/lib/slf4j-log4j12-1.7.10.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]

Logging initialized using configuration in file:/opt/hive/conf/hive-log4j2.properties Async: true
Hive-on-MR is deprecated in Hive 2 and may not be available in the future versions. Consider using a different execution engine (i.e. spark, tez) or using Hive 1.X releases.

hive> CREATE EXTERNAL TABLE IF NOT EXISTS weather_data (
>   date STRING,
>   temperature_2m_max DOUBLE,
>   temperature_2m_min DOUBLE,
>   city STRING
> )
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE
> LOCATION '/data/weather';
```



weather_data

DATABASE
 presto Presto

SCHEMA
 default

TABLE
 weather_data

weather_data

Table columns

Column Name	Datatype
date	VARCHAR
temperature_2m_max	FLOAT
temperature_2m_min	FLOAT
city	VARCHAR

CANCEL CREATE DATASET AND CREATE CHART

Create a new chart

✓ Choose a dataset

weather_data Add a dataset or view instructions

2 Choose chart type

Search all charts

All charts

Featured

Category

- Correlation
- Distribution
- Evolution
- Flow
- KPI
- Map
- Part of a Whole
- Ranking
- Table
- Other

Tags

Area Chart

Bar Chart

Big Number

Big Number with Trendline

Box Plot

Bubble Chart

Funnel Chart

Gauge Chart

Graph Chart

Heatmap

Line Chart

Map Chart

Pie Chart

Pivot Table

Radar Chart

Scatter Plot

Sunburst Chart

Table

Tree Chart

Treemap

Waterfall Chart

World Map

Please select both a Dataset and a Chart type to proceed CREATE NEW CHART

Add the name of the chart

SAVE

Chart Source: default.weather_data

Search Metrics & Columns

Metrics: Showing 1 of 1
f(x) COUNT(*)

Columns: Showing 4 of 4
abc date
temperature_2m_max
temperature_2m_min
abc city

DATA

LINE CHART

Query

X-AXIS
x abc date

X-AXIS SORT BY
Category name

X-AXIS SORT ASCENDING

METRICS
f(x) AVG(temperature_2m_max)

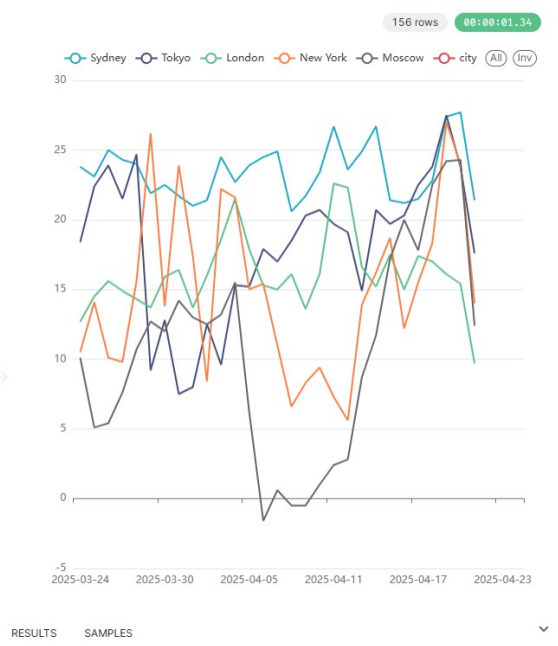
DIMENSIONS
x abc city

CONTRIBUTION MODE
Select ...

FILTERS
+ Drop columns/metrics here or click

SERIES LIMIT

UPDATE CHART



Superset Dashboards Charts Datasets SQL + Settings

Add the name of the chart

SAVE

Chart Source: default.weather_data

Search Metrics & Columns

Metrics: Showing 1 of 1
f(x) COUNT(*)

Columns: Showing 4 of 4
abc date
temperature_2m_max
temperature_2m_min
abc city

Save chart

Save (Overwrite) Save as...

CHART NAME*
weather_data_line_chart

ADD TO DASHBOARD
[untitled dashboard]

CANCEL SAVE & GO TO DASHBOARD SAVE

METRICS
f(x) AVG(temperature_2m_max)

DIMENSIONS
x abc city

CONTRIBUTION MODE
Select ...

FILTERS
+ Drop columns/metrics here or click

SERIES LIMIT

UPDATE CHART

156 rows 00:00:01.34

New York Moscow city (All) (Inv)

RESULTS SAMPLES

weather_data

☆ Draft

Superset Admin 17 minutes ago

EDIT DASHBOARD

...

weather_data_line_chart

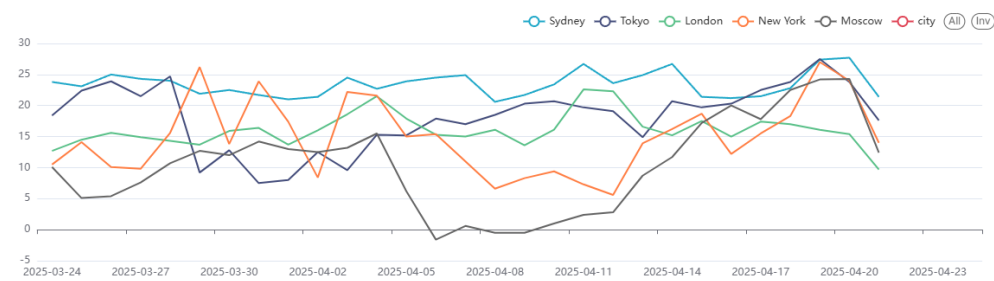


weather_data ☆ Draft Superset Admin 17 minutes ago

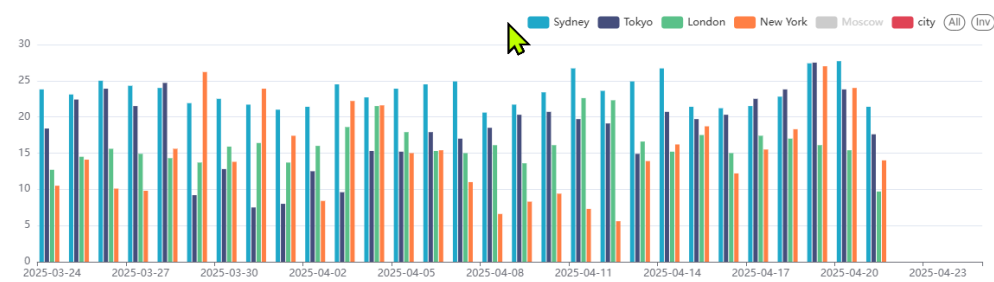
EDIT DASHBOARD

...

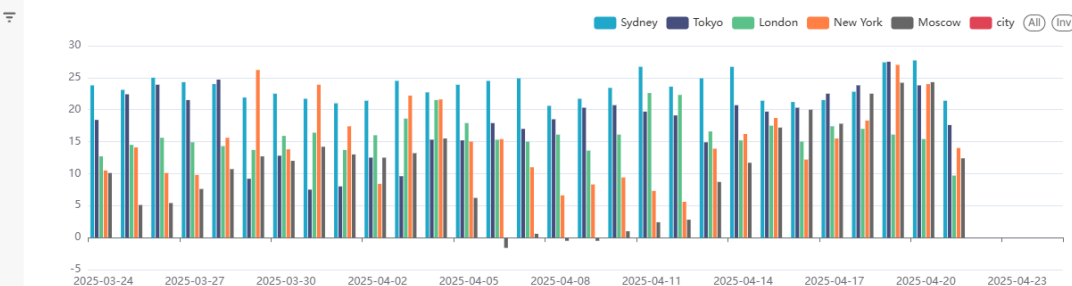
weather_data_line_chart



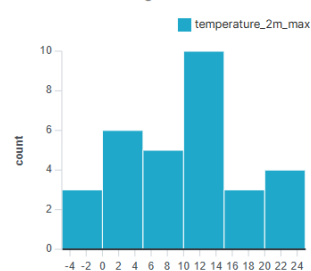
weather_data_bar_chart



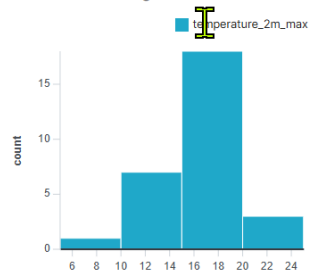
weather_data_bar_chart



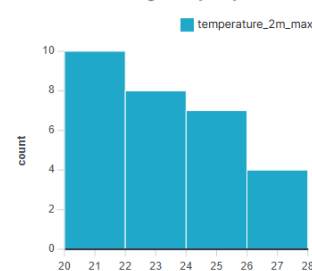
weather_data_histogram_Moscow



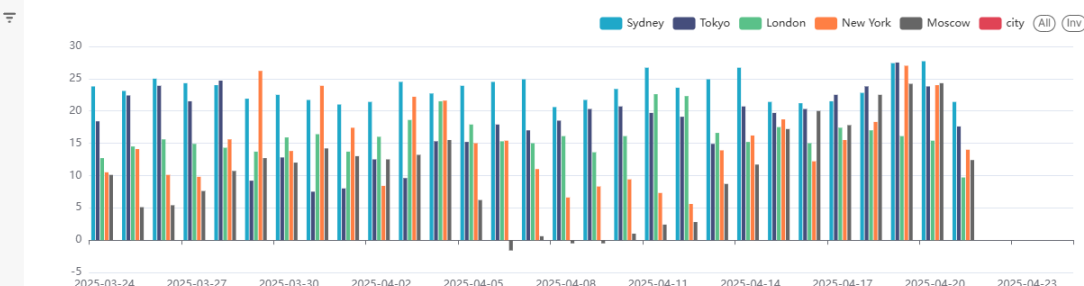
weather_data_histogram_London



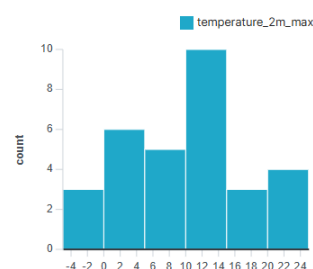
weather_data_histogram_Sydney



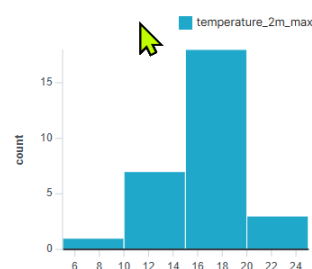
weather_data_bar_chart



weather_data_histogram_Moscow



weather_data_histogram_London



weather_data_histogram_Sydney

